

6	(a) p.d. = work (done) / charge OR energy transferred from (electrical to other forms) / (unit) charge	B1	[1]
(b)	(i)	$R = \rho l / A$	C1
		$\rho = 18 \times 10^{-9}$	C1
		$R = (18 \times 10^{-9} \times 75) / 2.5 \times 10^{-6} = 0.54 \, \Omega$	A1
			[3]
	(ii)	$V = IR$	C1
		$R = 38 + (2 \times 0.54)$	C1
		$I = 240 / 39.08 = 6.1 \, (6.14) \, \text{A}$	A1
			[3]

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(iii) $P = I^2 R$ or $P = VI$ and $V = IR$ or $P = V^2/R$ and $V = IR$ C1
 $= (6.14)^2 \times 2 \times 0.54$ C1
 $= 41 \text{ (40.7) W}$ A1 [3]

(c) area of wire is less (1/5) hence resistance greater ($\times 5$) M1
OR R is $\propto 1/A$ therefore R is greater
p.d. across wires greater so power loss in cables increases A1 [2]